skip to main content

# Quiz Submissions - CS571 Spring 2022: Web Technologies Exam 1

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## Attempt 1

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## Submission View

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## JSON Questions

**Question 1** **2 / 2 points**

[2 Points] Consider the following example from w3schools.com.

```
<!DOCTYPE html>
<html>
<body>

<p id="demo"></p>

<script>
const myJSON = '{"name":"John", "age":30, "cars":["Ford", "BMW", "Fiat"]}';
const myObj = JSON.parse(myJSON);

document.getElementById("demo").innerHTML = myObj.cars[1];
</script>

</body>
</html>
```

What string is displayed in the browser window?

○ A) Ford

○ B) <nothing>

● C) BMW

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**Question 2** **0 / 2 points**

[2 Points] Consider the following example from w3schools.com.

```
<!DOCTYPE html>
<html>
<body>
<p id="demo"></p>
<script>
const myJSON = '{"name":"John", "age":30, "car":null}';
const myObj = JSON.parse(myJSON);
let text = "";
for (const x in myObj) {
  text += myObj[x] + ", ";
}
document.getElementById("demo").innerHTML = text;
</script>
</body>
</html>
```

What string is displayed in the browser window?

Correct Answer ○ A) John, 30, null,

Incorrect Response ● B) <a syntax error>

○ C) name, age, car,

○ D) John, 30, null

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**Question 3** **1.667 / 2 points**

[2 Points] Select all statements that are true of JSON. Select all that apply.

Correct Answer ☑ JSON data can easily be sent between computers, and used by any programming language

Correct Answer ☑ JavaScript has a built in function for converting JSON strings into JavaScript objects

Correct Answer ☑ JavaScript has a built in function for converting a JavaScript object into a JSON string

Correct Answer ☑ The JSON format is syntactically similar to the code for creating JavaScript objects

Incorrect Response ☑ JavaScript has a built in function for converting an XML object into a JSON string false

Correct Answer ☑ You can receive pure text from a server and use it as a JavaScript object

**Question 4** **2 / 2 points**

**[2 Points] Which of the following are true statements for JSON? Select all that apply.**

☑ JSON is plain text written in JavaScript object notation true

☑ **Subset of ECMA-232 Third Edition** It's ECMA-262 3rd edition!

☑ **Language independent** true

☑ JSON is a lightweight data-interchange format true

**Question 5** **1 / 1 point**

[1 Point] A JSON encoder must only produce well-formed JSON text.

● True
○ False

**Question 6** **0 / 1 point**

[1 Point] JSON names require double quotes.

Correct Answer ○ A) True
Incorrect Response ● B) False

**Question 7** **1 / 1 point**

[1 Point] JSON is like XML because ... Select all that apply.

☐ A) Both use end tags

☐ B) Both require similar bandwidth JSON (JavaScript Object Notation) uses less bandwidth than XML because JSON files are smaller. This makes JSON a better choice for applications that need to transmit data quickly, like mobile apps and web services

☑ C) Both JSON and XML are "self describing" (human readable)

☑ D) Both JSON and XML are hierarchical (values within values)

**Question 8** **1 / 1 point**

[1 Point] The drawback of the eval() function is overcome with JSON.parse().

● A) True
○ B) False

**Question 9** **1 / 1 point**

[1 Point] The first index in a JSON array is zero (0).

○ True
● False

**Question 10** **0 / 1 point**

[1 Point] JSON Arrays are an unordered sequence of values, comma-separated and enclosed in square brackets

Incorrect Response ● A) True
Correct Answer ○ B) False

json.org specifies that "An object is an unordered [emphasis mine] set of name/value pairs", whereas an array is an "ordered collection of values". In other words

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## HTTP Questions

**Question 11** **1.333 / 2 points**

[2 Points] Which of the following are valid values of the Accept-Encoding header? Select all that apply.

Correct Answer ☑ deflate

Incorrect Response ☑ zlib zlib不是！！

Correct Answer ☑ gzip

**Question 12** **1 / 2 points**

[2 Points] How does a server send the type of the information it is providing to the browser? Select all that apply.

Correct Answer
Incorrect Response ☐ MIME media type

Correct Answer ☑ Content-Type header

**Question 13** **2 / 2 points**

[2 Points] What is the purpose of the HSTS header? Strict-Transport-Security (HSTS) Force communication using HTTPS instead of HTTP.

○ A performance feature that improves caching of web pages

Cache-Control
Directives for caching mechanisms in both requests and responses.

● A security feature that lets a web site tell browsers that it should only be communicated with using HTTPS

○ A feature used to indicate whether or not a browser should be allowed to render a page in a <frame> or <iframe>.

X-Frame-Options (XFO)
Indicates whether a browser should be allowed to render a page in a <frame>, <iframe>, <embed> or <object>.

**Question 14** **1.333 / 2 points**

[2 Points] What are properties of ETags? Select all that apply.

Incorrect Response ☑ Similar to 304 Not Modified

Correct Answer ☑ Used in cache validation

Correct Answer ☑ Similar to fingerprints

**Question 15** **1.333 / 2 points**

[2 Points] Select any header that is used to improve cache performance. Select all that apply.

Correct Answer ☑ Cache-control

Correct Answer ☑ Expires

Incorrect Response ☑ Last-seen

**Question 16** **0 / 1 point**

[1 Point] The client (browser) sends the "Content-Encoding" header.

Incorrect Response ● A) True
Correct Answer ○ B) False

**Question 17** **0 / 1 point**

[1 Point] "Edge" servers in a n-tier architecture, usually serve HTML, CSS, JavaScript, PNG images from the "edge" of the internet.

Correct Answer ○ A) True
Incorrect Response ● B) False

## CSS Questions

**Question 18** **1 / 1 point**

[1 Point] Which of the following correctly specifies how a CSS tag can be referenced in an HTML element?

○ A) # to refer to the class attribute of an HTML tag
. to refer to the id attribute of an HTML tag
nothing to refer to the name of an HTML tag

○ B) . to refer to the class attribute of an HTML tag
nothing to refer to the id attribute of an HTML tag
# to refer to the name of an HTML tag

● C) . to refer to the class attribute of an HTML tag
# to refer to the id attribute of an HTML tag
nothing to refer to the name of an HTML tag

○ D) nothing to refer to the class attribute of an HTML tag
# to refer to the id attribute of an HTML tag
. to refer to the name of an HTML tag

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**Question 19** **1 / 1 point**

[1 Point] How does one select an HTML element with ID called `top` in CSS?

○ A) `.top{ }`

○ B) `top{ }`

○ C) `ID.top{ }`

● D) `#top{ }`

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**Question 20** **1.5 / 2 points**

[2 Points] Consider the following example from w3schools.com.

```
<!DOCTYPE html>
<html>
<head>
<style>
p.center {
  text-align: center;
  color: red;
}

p.large {
  font-size: 300%;
}
</style>
</head>
<body>

<h1 class="center">This is heading 1.</h1>
<p class="center">This is paragraph 1.</p>
<p class="center large">This is paragraph 2.</p>

</body>
</html>
```

Select all the actual behaviors of this code. Select all that apply.

Correct Answer ☑ *paragraph 2* will be red, center-aligned, and in a large font-size

Correct Answer
Incorrect Response ☐ *heading 1* is not affected by listed the CSS style

Correct Answer ☑ *paragraph 1* will be red and center-aligned

☐ *heading 1* will be centered

**Question 21** **2 / 2 points**

[2 Points] Consider the following example from w3schools.com.

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  width: 100px;
  height: 100px;
  background-color: red;
  animation-name: example;
  animation-duration: 4s;
}

@keyframes example {
  from {background-color: red;}
  to {background-color: yellow;}
}
</style>
</head>
<body>

<h1>CSS Animation</h1>

<div></div>

</body>
</html>
```

What happens when the animation is finished? Select all that apply.

☐ Square is yellow

☐ Style is set to last frame

Default Behavior:
The animation does not persist the final state (background-color: yellow).
By default, the styles revert to the original state (background-color: red) after the animation ends.

☑ It goes back to the original style

☑ Square is red

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**Question 22** **2 / 2 points**

[2 Points] Which of the following are true of CSS? Select all that apply.

☐ few websites use external CSS style sheets

☐ CSS properties are not inherited from parent elements

☑ .class rule can be applied to multiple elements

**Question 23** **2 / 2 points**

[2 Points] Consider the following CSS.

```
.showNav ul li a:hover{color:#202022;}
```

And the following HTML snippet.

```
<div class="showNav">
     <ul>
          <li>
              <SOME HTML>
          </li>
     </ul>
</div>
```

---

Which of these examples to replace <SOME HTML> will trigger the CSS?

○ `<b> some navigation </b>`

● `<a href="#label">go there</a>`

○ `<img href="pict.png">`

## HTML DOM Questions

**Question 24** **1 / 1 point**

[1 Point] The DOM is available in Java and JavaScript implementations

● True
○ False

**Question 25** **1 / 1 point**

[1 Point] In HTML, which element do we use to create an unordered list?

○ <ol>

○ <li>

● <ul>

○ <unordered_list>

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**Question 26** **0 / 1 point**

[1 Point] What is the "novel" idea that Tim Berners-Lee invented?

Correct Answer ○ Hyperlinking

○ Hypercard

Incorrect Response ● HTML and CSS

○ Deep Linking

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**Question 27** **0 / 1 point**

[1 Point] What are required components of a web URL?

○ A) protocol, host, port, path

Incorrect Response ● B) protocol, host, port

○ C) protocol, host, port, path, query

Correct Answer ○ D) protocol, host

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**Question 28** **2 / 2 points**

[2 Points] Consider the following example from w3schools.com.

```
<!DOCTYPE html>
<html>
<head>
<style>
.democlass {
color: #ff0000;
}
</style>
</head>
<body>

<h1>Hello World</h1>
<button onclick="myFunction()">Try it</button>
<script>
function myFunction() {
var h1 = document.getElementsByTagName("H1")[0];
var att = document.createAttribute("class");
att.value = "democlass";
h1.setAttributeNode(att);
}
</script>
</body>
</html>
```

What happens when one clicks the "Try it" button?

○ Changes the color of "Hello World" to Green.

● Changes the color of "Hello World" to red.

○ Creates a "id" attribute and insert it to the H1 element above.

○ Adds a label "democlass" and colors it red.

**Question 29** **0 / 2 points**

[2 Points] Consider the following example from w3schools.com.

```
<!DOCTYPE html>
<html>
<body>
<p>Click the button.</p>
<button onclick="myFunction()">Try it</button>
<script>
function myFunction() {
var btn = document.createElement("BUTTON");
btn.innerHTML = "CLICK ME";
document.body.appendChild(btn);
}
</script>
</body>
</html>
```

What happens when one clicks the "Try it" button?

Incorrect Response ● Creates a BUTTON element with text "CLICK ME" below (i.e., under) the "Try It" button

○ Creates a BUTTON element with text "CLICK ME" above the "Try It" button

○ Creates a BUTTON element with text "CLICK ME" above the text "Click the button".

Correct Answer ○ Creates a BUTTON element with text "CLICK ME" next to the "Try It" button

**Question 30** **2 / 2 points**

[2 Points] Consider the following example from w3schools.com.

```
<!DOCTYPE html>
<html>
<body>

<p>This example uses the addEventListener() method to attach a click event to the document.</p>

<p>Click anywhere in the document.</p>

<p id="demo"></p>

<script>
document.addEventListener("click", function(){
  document.getElementById("demo").innerHTML = "Hello World!";
});
</script>

</body>
</html>
```

What happens when one clicks TWICE anywhere in the document ?

● "Hello World!" is printed below the Note

○ "Hello World!" is printed above the Note

○ Nothing

**Question 31** **0 / 2 points**

[2 Points] Consider the following example from w3schools.com.

```
<!DOCTYPE html>
<html>
<body>

<h2 class="example">A heading with class="example"</h2>
<p class="example">A paragraph with class="example".</p>

<p>Click the button to add a background color to the first element in the document with class="example".</p>

<button onclick="myFunction()">Try it</button>

<script>
function myFunction() {
  document.querySelector(".example").style.backgroundColor = "red";
}
</script>

</body>
</html>
```

What happens when one clicks the "Try it" button?

○ Adds red background color to the paragraph with text *A paragraph with class="example"*.

Correct Answer ○ Adds red background color to the heading with text *A heading with class="example"*

Incorrect Response ● Adds red background color to both the header and the paragraph

**Question 32** **2 / 2 points**

[2 Points] Consider the following example from w3schools.com.

```
<!DOCTYPE html>
<html>
<body>

<button onclick="removeHandler()">Try it</button>

<p id="demo"></p>

<script>
document.addEventListener("mousemove", myFunction);

function myFunction() {
  document.getElementById("demo").innerHTML = Math.random();
}

function removeHandler() {
  document.removeEventListener("mousemove", myFunction);
}
</script>

</body>
</html>
```

What happens when one clicks the "Try it" button?

○ A random number is displayed every time you move your mouse in this document.

● Nothing

○ A random number is displayed immediately

## JavaScript / RegEx Questions

**Question 33** **2 / 2 points**

[2 Points] Using document.write() after an HTML document is loaded, will **delete all existing HTML**.

● a) True
○ b) False

**Question 34** **2 / 2 points**

[2 Points] Consider the following example from w3schools.com.

```
<!DOCTYPE html>
<html>
<body>

<p id="demo"></p>

<script>
let str = "Apple, Banana, Kiwi";
document.getElementById("demo").innerHTML = str.slice(7,13);
</script>

</body>
</html>
```

What output will be displayed on the page?

○ A) anana,

● B) Banana

○ C) , Banan

○ D) Nothing

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**Question 35** **2 / 2 points**

[ 2 Points] Consider the following JavaScript statements:

```
foo = 5;
bar = "5";
num1 = foo + bar;
```

---

What will be value of num1?

○ 55 (numeric)

● "55" (string)

**Question 36** **2 / 2 points**

[2 Points] Consider the following program:

```
var re = /\w+\s/g;
var str = "fee fi fo fum";
var myArray = str.match(re);
console.log(myArray);
```

---

What output is displayed on the console?

● Array(3) [ "fee ", "fi ", "fo " ]

○ Array(4) [ "fee ", "fi ", "fo ", "fum" ]

**Question 37** **2 / 2 points**

[2 Points] Consider the following example from w3schools.com.

```
<!DOCTYPE html>
<html>
<body>

<h2>My First Web Page</h2>
<p>My first paragraph.</p>

<button type="button" onclick="document.write(5 + 6)">Try it</button>

</body>
</html>
```

What output will be displayed on the page after clicking the **Try it** button?

● A) 11

○ B) My First Web Page, My first paragraph, Try it button

○ C) Nothing, the screen is blank

○ D) My First Web Page, My first paragraph, Try it button, 11

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**Question 38** **2 / 2 points**

[2 Points] Consider the following example from w3schools.com.>

```
<!DOCTYPE html>
<html>
<body>

<p id="demo"></p>

<script>
let str = "Apple, Banana, Kiwi";
document.getElementById("demo").innerHTML = str.substring(7,13);
</script>

</body>
</html>
```

What string is displayed in the browser window?

○ A) anana<space>

○ B) <space<Banana>

○ C) <space>Banan

● D) Banana

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**Question 39** **1.5 / 2 points**

[2 Points] Consider the following guidelines for a security policy for passwords:

- at least *8* characters
- combination of upper- and lower-case characters
- one or more digits
- not a dictionary word

Select all that apply.

Correct Answer ☑ They are no longer considered secure

☐ They provide strong security

☐ They are better than using Diceware

Correct Answer
Incorrect Response ☐ Should be avoided

## Networking Questions

**Question 40** **1.333 / 2 points**

[2 Points] What is TRUE of SpaceX Starlink? Select all that apply.

Correct Answer
Incorrect Response ☐ A constellation

Correct Answer ☑ It is implemented using satellites at various orbits

Correct Answer ☑ Provides Internet service

**Question 41** **2 / 2 points**

[2 Points] What is TRUE of NAT (Network Address Translation)? Select all that apply.

☑ It allows LAN computers to have a private IP address

☑ The NAT server is usually implemented in a router

☑ Allows every computer on the Local Area Network (LAN) to send out packets with the same IP address but different source port

☐ The NAT server's private IP address has been substituted with the computer's public IP address in a packet

**Question 42** **2 / 2 points**

[2 Points] Ports from 0-1023 (inclusively) can be used by:

○ Any application

○ The operating system when an application needs to pass an application off to a non-registered port

○ People who enjoy dessert wines

● Well-known applications, so root or administrator access is required

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**Question 43** **2 / 2 points**

[2 Points] What is TRUE of the following terminal snapshot? Select all that apply.
1 192.168.7.1 (192.168.7.1) 5.077 ms 2.879 ms 2.026 ms 2 142-254-186-181.inf.spectrum.com (142.254.186.181) 9.374 ms 12.567 ms 12.720 ms 3 76.167.30.154 (76.167.30.154) 14.682 ms 11.227

☑ It can be used to find the IP routing between source and destination

☑ '*' means request timed out after 5 seconds

☐ It is visible only to system administrators

☑ It displays running `tracert` on Windows or `traceroute` on macOS

**Question 44** **2 / 2 points**

[2 Points] Is 172.21.101.211 a public IP Address or a private IP Address?

○ Both

● Private

○ Public

○ Neither

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## Python Questions

**Question 45** **2 / 2 points**

[2 Points] Consider the following Python code:

```
>>> stack = [3, 4, 5]
>>> stack.append(6)
>>> stack.append(7)
>>> stack.pop()
>>> stack
```

What gets printed?

● [3, 4, 5, 6]

○ [3, 4, 6, 7]

○ [4, 5, 6, 7]

**Question 46** **2 / 2 points**

[2 Points] Consider the following Python code:

```
thislist = ["apple", "banana", "cherry", "orange", "kiwi", "melon", "mango"]
print(thislist[:4])
```

What gets printed?

● A) ['apple', 'banana', 'cherry', 'orange']

○ B) ['apple', 'banana', 'cherry'] 不包括index4

○ C) ['apple', 'banana', 'cherry', 'orange', 'kiwi']

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**Question 47** **2 / 2 points**

[2 Points] Consider the following Python code:

```
thislist = ["apple", "banana", "cherry", "orange", "kiwi", "melon", "mango"]
print(thislist[-4:-1])
```

What gets printed?

○ A) ['orange', 'kiwi', 'melon', 'mango']

● B) ['orange', 'kiwi', 'melon']

○ C) ['kiwi', 'melon']

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**Question 48** **2 / 2 points**

[2 Points] Consider the following Python code:

```
x = {"name" : "John", "age" : 36}

#display x:
print(x)

#display the data type of x:
print(type(x))
```

What gets printed?

○ {'name': 'John', 'age': 36}
<class 'set'>

● {'name': 'John', 'age': 36}
<class 'dict'>

○ {'name': 'John', 'age': 36}
<class 'tuple'>

**Question 49** **2 / 2 points**

[2 Points] Consider the following Python code:

```
thislist = ["apple", "banana", "cherry", "orange", "kiwi", "melon", "mango"]
print(thislist[2:5])
```

What gets printed?

● A) ['cherry', 'orange', 'kiwi']

○ B) ['banana', 'cherry', 'orange']

○ C) ['cherry', 'orange', 'kiwi', 'melon']

View Feedback

**Question 50** **0.75 / 1 point**

[1 Point] Which of the following is **TRUE** for Python "sets"? Select all that apply.

Correct Answer
Incorrect Response ☐ A) A set is a collection which is *unchangeable*, where set items are unchangeable, but you can remove items and add new items

☐ B) A set is a collection which is *ordered*

☐ C) A set is a collection which is *indexed*

Correct Answer ☑ D) Sets are used to store multiple items in a single variable.

View Feedback

**Question 51** **1 / 1 point**

[1 Point] Which of the following is **TRUE** for Python "dictionaries"? Select all that apply.

☑ A) Dictionary items can be removed after the dictionary has been created

☑ B) A dictionary is a collection which is *changeable*

☑ C) Dictionaries are used to store data values in key:value pairs

☐ D) Duplicate dictionary items are allowed

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**Question 52** **1 / 1 point**

[1 Point] In Python, the data type is set when you assign a value to a variable.

● A) True
○ B) False

**Question 53** **1 / 1 point**

[1 Point] Which of the following is **TRUE** for Python "**tuples**"? Select all that apply.

☐ A) Tuple items are indexed, the first item has index [1]

☑ B) We cannot change, add or remove items after the tuple has been created

☑ C) The order of tuple items will not change

☐ D) Tuple items cannot have the same value

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**Question 54** **1 / 1 point**

[1 Point] Which of the following is **TRUE** in Python? Select all that apply.

☑ A) Variables that are created outside of a function are known as "global" variables.

☐ B) Global variables must start with an underscore.

☐ C) Global variables should use only uppercase letters.

☑ D) Global variables can be used by everyone, both inside of functions and outside.

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**Question 55** **1 / 1 point**

[1 Point] Which of the following is **TRUE** in Python? Select all that apply.

☐ A) Python has the 'var' command for declaring a variable.

☑ B) A variable is created the moment you first assign a value to it.

☑ C) Variables are containers for storing data values.

View Feedback

## HTML Questions

**Question 56** **1.333 / 2 points**

[2 Points] Select characters that to appear in HTML need to be represented with numeric or entity references. Select all that apply.

Correct Answer ☑ & (ampersand)

Correct Answer ☑ " (double quote)

Correct Answer
Incorrect Response ☐ ' (single quote)

**Question 57** **2 / 2 points**

[2 Points] When using Universal Resource Identifiers (URIs) in HTML to reference other HTML pages at different web sites, what are the **required** components of the URIs? Please select all that apply.

☑ scheme

☑ host name

**Question 58** **2 / 2 points**

[2 Points] Select all the ways to specify the URL in the src attribute of the <IMG> tag. Select all that apply.

☑ Relative URL

☑ Absolute URL

☐ Partial URL

☐ Embed URL

**Question 59** **2 / 2 points**

[2 Points] Consider the following HTML code sample.

```
<!DOCTYPE html>
<html>
<body>

<img src="img_typo.jpg" alt="Girl with a jacket">

</body>
</html
```

---

What is displayed on the web page, if the image `img_type.jpg` does not exist?

○ Nothing

● Girl with a jacket

**Question 60** **2 / 2 points**

[2 Points] Consider the following statement:

HTML parsers are very forgiving and throw errors only in extreme conditions.

○ a) True
● b) False

---

**Attempt Score:**
82.08 / 100 - 82.08 %
**Overall Grade** (highest attempt)**:**
82.08 / 100 - 82.08 %

Done